

landscape or in portrait mode. So, here, contact gesture recognition plays an important role.

Accelerometers have paved the way for many games to be developed for smartphones, especially for some of those sports simulation games that let you use the phone as a steering wheel or a fishing pole. To get an idea of the range of accelerometers that you can select from, refer to Table I.

Gyroscopes. Gyroscopes, also popularly known as gyros, have had a major impact in the world of gesture recognition. Their ability to measure the orientation of a device enables them to measure both vertical and horizontal rotation. This has brought smoothness and preciseness to the operation of a mobile device. The next time you hold a phone on its side, and the screen transforms into landscape mode automatically, you know which component to thank.

Application. In the case of a smartphone or a tablet, adding a gyro and an accelerometer allows the device to sense motion on six axes including roll, pitch and yaw—which helps you fly your airplane in a flight simulator game on a phone. So if accelerometers have given more business to game developers, gyros have helped them be more innovative.

Bend sensors. These are also known as flex sensors. As we know, even finger movements are considered as gestures, so bend sensors help us to identify those gestures by recognising finger postures. They work on the principle that resistance increases with an increase in the bend.

Data gloves. Gesture recognition technology has not only been implemented in smartphones and tablets, but also in television sets and computers. In order to implement hand gesture recognition on these devices, data gloves have been developed. These gloves identify the finger and hand gestures in real time scenario. These are wired either to a TV set or to a computer and then worn by the person operating the device. These act

as an input device and eliminate the use of a mouse or a keyboard.

Application. The bend sensors are placed on the joints of all the fingers in the data glove, so the more the fingers are bent, the more is the resistance which is introduced in the device for further action.

Infrared proximity sensors. These have the ability to detect the presence of nearby objects without any physical contact. They emit an electromagnetic field and then identify the changes either in the field or in the return signal, which they detect after reflection from an object that has got close to the sensor. There are many types of proximity sensors; however, infrared proximity sensors play a very important role in gesture recognition. As soon as these infrared sensors recognise the movement of a body in front of them, they pass a signal to the device for further action.

Application. Many of us have seen a wash basin with an automated tap. The tap turns on as soon as it senses some hand movement under it. This is because of the presence of infrared proximity sensors that detect your hand movement. Here's a fun thing to try out—if you keep your hand under the tap without any movement, the tap would not allow the water to fall. This is because if you keep your hand immobile, the sensor does not sense a changing field and thus thinks that your hand is not in front of it any more.

Capacitive proximity sensors. They support gesture recognition enabling robust, easy-to-use and feature-rich user touch interfaces on a device—these are what make up your capacitive touch screen. Such sensors allow users to operate the device just by the touch of their skin or finger rather than by applying pressure. Some of the newer models have superior sensors that do not even require a touch—hovering above the screen is enough to get their attention.

Application. This has helped to bring about a new era of devices with superior touch screens that allow mul-



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Tesca Technologies Pvt. Ltd.

Office : 305, Taru Chhaya Nagar, Tonk Road, Jaipur-302029, India

Factory : IT-2013, Ramchandrapura Indl. Area, Sitapura, Jaipur, India.

Phone : +91-141-2724326 ; +91-9413330765

Email : info@tesca.in,
tesca.technologies@gmail.com

Web : www.tesca.in